

Software Requirements Specifications for
ThriftFinder: A Web-Based Secondhand Marketplace and Reservation System



In Partial Fulfillment
Of the Requirements of the Subject
IT - 305 Integrative Programming and Technologies 2
Bachelor of Science in Information Technology

By:

Evedientes, Khim Jay T.

Petalino, Cyrus Azhly G.

TABLE OF CONTENTS

1. INTRODUCTION

1.1 PURPOSE

1.2 INTENDED AUDIENCE

1.3 PRODUCT SCOPE

1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

1.5 DOCUMENT CONVENTIONS

1.6 REFERENCES

2. PROJECT PROPOSAL AND TEAM FORMATION

3. OVERALL DESCRIPTION

3.1 PRODUCT PERSPECTIVE

3.2 USER CLASSES AND CHARACTERISTICS

3.3 OPERATING ENVIRONMENT

3.4 DESIGN AND IMPLEMENTATION CONSTRAINTS

3.5 ASSUMPTIONS AND DEPENDENCIES

4. REQUIREMENTS ANALYSIS

4.1 FUNCTIONAL REQUIREMENTS

4.2 NON-FUNCTIONAL REQUIREMENTS

4.3 REQUIREMENT PRIORITIZATION AND TRACEABILITY

4.4 BUSINESS RULES

4.5 EXTERNAL INTERFACE REQUIREMENTS

5. SYSTEM MODELS AND ARCHITECTURE

6. BACK-END DEVELOPMENT

7. FRONT-END AND BACK-END INTEGRATION

8. EXTERNAL API / TECHNOLOGY INTEGRATION

9. SECURITY IMPLEMENTATION

10. GUI PROTOTYPING

11. DATA AND DATABASE REQUIREMENTS

12. VERIFICATION, TESTING, AND DEBUGGING

13. DEPLOYMENT, FINALIZATION, AND DOCUMENTATION

14. APPENDICES

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for ThriftFinder, a web-based platform that allows users to list, browse, and reserve secondhand items. This document is intended to guide design, development, and testing for the capstone project and to serve as the agreed basis of scope between the development team and the capstone panel.

1.2 Intended Audience

- Development team members (implementation and testing reference)
- Capstone adviser and panel (scope and requirements review)
- Buyers and Sellers
- Future maintainers of the system

1.3 Product Scope

ThriftFinder is a responsive web application that connects sellers of secondhand goods with buyers within a local community. Sellers can list items for sale; buyers can browse, search, and place a time-limited reservation on an item before arranging an in-person handover. The system's core value is reducing wasted trips and double-booking by formalizing the "reserve before you go" step that is normally handled informally through chat groups or marketplace comment threads.

1.4 Definitions, Acronyms, and Abbreviations

To ensure consistent understanding of technical terminology throughout this document, the following definitions, acronyms, and abbreviations are provided:

- **ThriftFinder:** The proposed web-based secondhand marketplace and reservation system described in this document.
- **Listing:** An item posted for sale by a seller on ThriftFinder.
- **Reservation:** A time-limited hold placed by a buyer on a listing, pending confirmation and handover.
- **Holding Period:** The fixed window of time during which a reservation must be confirmed before it automatically expires.
- **Admin:** A platform user with moderation privileges over listings and user accounts.
- **Buyer:** A registered user who browses listings and places reservations on items.
- **Seller:** A registered user who creates and manages listings for items they wish to sell.
- **MVP:** Minimum Viable Product — the core set of High-priority features required for the system to function and be able to be demonstrated, as will be defined in Section 4.

1.5 Document Conventions

Functional requirements are labeled FR-#. Priority levels (High/Medium/Low) indicate implementation order: High-priority items form the Minimum Viable Product (MVP) described in Section 3.1; Medium and Low items are enhancements to be attempted only after the MVP is complete and stable.

1.6 References

- Aldon, D. K. M., Tan, C. S. L., & Islam, S. (2026). From necessity to identity: Entrepreneurial perspectives on key drivers of second-hand clothing consumption in the Philippines' ukay-ukay sector. *Frontiers in Sustainability*, 7, Article 1724156.
<https://doi.org/10.3389/frsus.2026.1724156>
- Abad, M. A. H., Reyes Jr, R. V., Duplon, E. J., Vallejo, A. P., Malilay, J. R., & Aduna, R. R. (2025). Cultural and social drivers of Ukay-Ukay shopping trends in Nueva Ecija: An analysis of consumer behavior. *International Journal for Multidisciplinary Research*.
<https://doi.org/10.62225/2583049X.2025.5.4.4570>
- Cabañero, R. A. C. (2023). Tiffany: Ukay-Ukay online shop. *International Journal of Advanced Research in Science, Communication and Technology*, 3(1).
<https://doi.org/10.48175/IJARSCT-12301>
- [Author]. (2023). Determinants affecting Ukay-Ukay (thrift clothes) business owners under the new normal in Barangay New Isabela, Tacurong City. St. Augustine's University.
- Systems Requirements Specification(SRS) Template
<https://classroom.google.com/u/1/c/ODcxNTIxNjg4ODU4>

2. Project Proposal and Team Formation

Project Title: ThriftFinder: A Web-Based Secondhand Marketplace and Reservation System

Proposed By: Khim Jay T. Evedientes, Cyrus Azhly G. Petalino

Instructor: Jaafar Omar

2.1 Approved Concept and Reason for Selection

ThriftFinder was selected from the team's pool of proposed concepts and formally approved by the capstone panel. The concept addresses a recurring, observable problem in local secondhand ("ukay-ukay") commerce: items are typically sold through informal channels such as Facebook groups or physical stalls, where buyers cannot reliably hold an item before traveling to see it in person. This results in wasted trips, items being sold to whoever arrives first regardless of prior online interest, and disputes over who "called dibs" on an item. The team selected this concept because it is scoped tightly enough to be buildable within one semester (Section 3.4), is grounded in documented local consumer behavior (Abad et al., 2025; Aldon et al., 2026 — see Section 1.6), and satisfies the course's technical checklist (CRUD, REST API, and related requirements) through a genuinely useful feature — the reservation system — rather than a feature bolted on solely to satisfy the rubric.

2.2 Team Members, Roles, and Responsibilities

Member	Assigned Role	Responsibilities
Khim Jay T. Evedientes	[e.g., Project Manager / Back-End Developer]	[e.g., API design, database, reservation logic, documentation]
Cyrus Azhly G. Petalino	[e.g., Front-End Developer / UI-UX Designer]	[e.g., React UI, GUI prototyping, front-end/back-end integration]

2.3 Problem Statement, Objectives, Target Users, Core Features, and Scope Boundaries

Problem Statement: Local buyers and sellers of secondhand goods currently rely on informal, unstructured channels (chat groups, walk-in stalls) with no reliable way to hold an item pending pickup, leading to wasted trips, double-booked items, and buyer/seller frustration.

Objectives:

- To provide sellers a structured way to list secondhand items with photos, price, and condition.
- To provide buyers a searchable, filterable catalog of available items.
- To let buyers place a time-limited reservation on an item before arranging in-person handover, reducing wasted trips and disputes.
- To give both parties a way to communicate and confirm handover details without leaving the platform.

Target Users: Buyers and sellers of secondhand goods within a local community (e.g., students and residents in Pagadian City / the Zamboanga Peninsula), and a platform Admin responsible for moderation.

Core Features: Account registration/login, listing creation and browsing, search/filtering, item reservation with automatic expiry, buyer–seller chat, notifications, ratings/reviews, and admin moderation — detailed fully as FR-001 through FR-012 in Section 4.1.

Scope Boundaries: The full in-scope/out-of-scope breakdown is defined in Section 4.3.1–4.3.3. In summary, ThriftFinder handles listing, discovery, and reservation; it deliberately does not handle payment processing, delivery logistics, or native mobile apps (see Section 4.3.3 for the complete list and rationale).

2.4 Comparison of Similar or Existing Systems

System	Strengths	Gap ThriftFinder Addresses
Facebook Marketplace / Groups	Very large existing user base; free and easy to post an item.	No reservation mechanism — items are sold on a first-come, first-served basis at pickup, regardless of who messaged first, causing wasted trips.
Shopee / Lazada (2nd-hand listings)	Established checkout, payment, and buyer-protection infrastructure.	Built for new/mass-market retail; heavier than needed for informal peer-to-peer secondhand selling, and charges seller fees.
Carousell / OLX-style apps	Purpose-built for secondhand peer-to-peer selling with chat built in.	Not localized to this community; still lacks a formal "hold this item for me" reservation state with an expiry.
Physical ukay-ukay stalls	Immediate, tactile inspection of items before buying.	No remote discovery at all — a buyer must physically visit to know what's available.

Across all compared systems, none combine local secondhand-market focus with an explicit, time-bound reservation state — this is the specific gap ThriftFinder's core feature set (FR-006 through FR-008) is designed to fill.

2.5 Proposal Approval and Supporting Evidence

The proposed ThriftFinder: A Web-Based Secondhand Marketplace and Reservation System will be presented to the project adviser and course instructor for review and approval. Suggestions and recommendations regarding the system's features, scope, target users, and overall design will be considered and incorporated into the succeeding stages of the project.

Records of the proposal presentation, adviser feedback, and approval will be maintained as supporting evidence and may be included in the Appendices of this SRS.

3. Overall Description

3.1 Product Perspective

ThriftFinder is a new, standalone web application. It is not a modification of an existing system. It consists of a front-end web client, a back-end application server exposing a REST API, and a relational database. It does not depend on any other in-house system.

3.2 User Classes and Characteristics

ID	Feature	Description	Priority
Buyer	Registered User	Browses listings, places and manages reservations. No technical expertise assumed.	Primary
Seller	Registered User	Creates and manages listings, confirms reservations. No technical expertise assumed.	Primary
Admin	Platform Staff	Moderates listings/users, views reports. Assumed to be a project member during capstone demo.	Secondary

3.3 Operating Environment

- Client: modern desktop and mobile web browsers (Chrome, Edge, Firefox) via responsive design
- Server / Hosting: Vercel (Hobby/free tier) for the front end and API routes. Hobby is free indefinitely with no credit card, but is limited to personal/non-commercial, single-developer projects, 100 GB bandwidth, and short-lived serverless functions (10s execution) — sufficient for a capstone demo, but not meant for a paying production product.
- Database & Backend Services: Supabase (free tier) — provides a hosted PostgreSQL database, built-in authentication, and file storage (for listing photos) in one free plan, which avoids needing to stand up a separate database server.
- Rationale: This pairing (Vercel + Supabase) is a common free-tier combination for capstone-scale web apps: Vercel handles hosting/deployment, Supabase handles data, auth, and image storage, and both offer generous no-cost usage limits for a project this size.

3.4 Design and Implementation Constraints

- Must be deliverable and demonstrable within the remaining capstone timeline
- Team size and skill set limit implementation to web technologies already in the team's stack
- No budget for paid third-party services (payment gateways, SMS APIs, paid hosting tiers)
- Must comply with the Philippine Data Privacy Act for handling user personal information

3.5 Assumptions and Dependencies

- Users have access to a device with a modern web browser and internet connection
- Item handover happens in person; the system does not verify item condition or completion of the physical exchange
- Email delivery for notifications depends on a third-party or free-tier email service
- The system assumes good-faith use; full fraud prevention is out of scope for this version (see Section 4)

4. Requirements Analysis

This section presents the full functional and non-functional requirements for ThriftFinder, how they are prioritized and traced, the business rules governing the platform's behavior, and the external interfaces the system must support.

4.1 Functional Requirements

The table below lists every functional requirement. High-priority items form the Minimum Viable Product (MVP) required for defense; Medium and Low items are second-pass enhancements attempted only after the MVP is complete and demonstrable.

ID	Feature	Description	Priority
FR-001	Account Registration & Login	Users register/log in as Buyer or Seller (or both) using email and password. Includes email verification and password reset.	High
FR-002	User Profile	Users maintain a profile with name, contact info, profile photo, and view their own listing/reservation history.	High

FR-003	Create/Edit/Delete Listing	Sellers post secondhand items with title, category, condition, price, description, and up to 5 photos. Sellers can edit or remove active listings.	High
FR-004	Browse & Search Listings	Buyers browse all active listings and search/filter by keyword, category, price range, and condition.	High
FR-005	Listing Detail View	Buyers view full listing details: photos, description, price, seller info, and reservation status.	High
FR-006	Reserve Item	A buyer reserves an available item for a limited holding period (e.g., 24-48 hours) pending pickup/payment arrangement.	High
FR-007	Reservation Status Tracking	Reservations move through defined states: Pending → Confirmed → Completed, or Cancelled / Expired. Both parties can view current status.	High
FR-008	Automatic Expiry	A reservation that is not confirmed within the holding period automatically expires and the listing becomes available again. Implementation note: checked lazily (i.e., evaluated against the current time whenever the listing/reservation is read or acted on) rather than via a scheduled cron job, since the free hosting tier's cron only runs once	Medium

		per day with imprecise timing — unsuitable for a time-sensitive expiry window.	
FR-009	Buyer–Seller Chat	Once a reservation is confirmed, a buyer and seller can message each other in a real-time in-app chat thread to arrange handover details. Implemented using Supabase Realtime (Postgres change subscriptions) — no separate chat service required, and free-tier limits (200 concurrent connections, 2M messages/month) comfortably cover capstone-scale usage.	Medium
FR-010	Notifications	Users receive in-app (and/or email) notifications for reservation created, confirmed, cancelled, or expired events.	Medium
FR-011	Ratings & Reviews	After a completed reservation, buyer and seller can rate/review each other.	Low
FR-012	Admin Moderation Panel	Admins can view/remove listings and users that violate platform rules, and view basic platform activity reports.	Medium

Expanded detail per requirement, including source, related module, and acceptance criteria:

FR-001 — Account Registration & Login *(Priority: High)*

Users register/log in as Buyer or Seller (or both) using email and password. Includes email verification and password reset.

Source: Core platform need — every downstream feature (listing, reserving, chatting) requires a known, authenticated user.

Related Module: Authentication Module

Acceptance Criteria: User can register with a valid email and password; unverified accounts cannot log in; incorrect credentials show an error; a password-reset email is sent and the link works within its validity window.

FR-002 — User Profile *(Priority: High)*

Users maintain a profile with name, contact info, profile photo, and view their own listing/reservation history.

Source: Buyers and sellers need to see who they're transacting with before committing to a reservation.

Related Module: User Management Module

Acceptance Criteria: A logged-in user can view and edit their name, contact info, and profile photo; their profile correctly lists their own past and active listings/reservations.

FR-003 — Create/Edit/Delete Listing *(Priority: High)*

Sellers post secondhand items with title, category, condition, price, description, and up to 5 photos. Sellers can edit or remove active listings.

Source: Core seller workflow required to populate the marketplace with items to browse.

Related Module: Listing Management Module

Acceptance Criteria: A seller can publish a listing with all required fields and at least one photo; edits persist after saving; a deleted listing no longer appears in browse or search results.

FR-004 — Browse & Search Listings *(Priority: High)*

Buyers browse all active listings and search/filter by keyword, category, price range, and condition.

Source: Core buyer discovery workflow — buyers cannot reserve what they cannot find.

Related Module: Listing Management Module

Acceptance Criteria: The browse page lists all active listings; a keyword search returns only matching results; category, price-range, and condition filters each correctly narrow the result set.

FR-005 — Listing Detail View *(Priority: High)*

Buyers view full listing details: photos, description, price, seller info, and reservation status.

Source: Buyers need full information before deciding to reserve.

Related Module: Listing Management Module

Acceptance Criteria: Clicking any listing opens a detail page showing all uploaded photos, the full description, price, seller name, and the listing's current reservation status.

FR-006 — Reserve Item *(Priority: High)*

A buyer reserves an available item for a limited holding period (e.g., 24-48 hours) pending pickup/payment arrangement.

Source: Directly implements the platform's core value proposition (Section 1.3): formalizing "reserve before you go."

Related Module: Reservation Module

Acceptance Criteria: A buyer can reserve any listing not already under an active reservation; the listing immediately shows as reserved to other buyers; the reservation stores a correct holding-period deadline.

FR-007 — Reservation Status Tracking *(Priority: High)*

Reservations move through defined states: Pending → Confirmed → Completed, or Cancelled / Expired. Both parties can view current status.

Source: Both buyer and seller need shared visibility into where a reservation stands.

Related Module: Reservation Module

Acceptance Criteria: Both the buyer and seller dashboards display the same, correct current state for any given reservation at all times.

FR-008 — Automatic Expiry *(Priority: Medium)*

A reservation that is not confirmed within the holding period automatically expires and the listing becomes available again. Implementation note: checked lazily (i.e., evaluated against the current time whenever the listing/reservation is read or acted on) rather than via a scheduled cron job, since the free hosting tier's cron only runs once per day with imprecise timing — unsuitable for a time-sensitive expiry window.

Source: Prevents a listing from being stuck "reserved" indefinitely if a buyer never follows through.

Related Module: Reservation Module

Acceptance Criteria: A Pending reservation whose deadline has passed is shown as Expired the next time it is read, and the listing can be reserved by someone else.

FR-009 — Buyer–Seller Chat *(Priority: Medium)*

Once a reservation is confirmed, a buyer and seller can message each other in a real-time in-app chat thread to arrange handover details. Implemented using Supabase Realtime (Postgres change subscriptions) — no separate chat service required, and free-tier limits (200 concurrent connections, 2M messages/month) comfortably cover capstone-scale usage.

Source: Requested enhancement to replace informal off-platform coordination (e.g., separate messaging apps) for arranging handover.

Related Module: Communication Module

Acceptance Criteria: After a reservation is confirmed, both parties can send and receive messages in the same thread, and new messages appear without a page refresh.

FR-010 — Notifications *(Priority: Medium)*

Users receive in-app (and/or email) notifications for reservation created, confirmed, cancelled, or expired events.

Source: Keeps both parties informed without requiring them to actively poll the app for status changes.

Related Module: Communication Module

Acceptance Criteria: A notification is generated for each of the four listed reservation events and is visible to the affected user within a reasonable time.

FR-011 — Ratings & Reviews *(Priority: Low)*

After a completed reservation, buyer and seller can rate/review each other.

Source: Builds trust and accountability between anonymous marketplace users, consistent with findings in Chen et al. (2018) on secondhand seller reputation.

Related Module: Reputation Module

Acceptance Criteria: After a reservation reaches Completed, both parties can submit a rating and short review, which then displays on the reviewed user's profile.

FR-012 — Admin Moderation Panel (*Priority: Medium*)

Admins can view/remove listings and users that violate platform rules, and view basic platform activity reports.

Source: Required for basic platform governance and to demonstrate the moderation/admin role during the defense.

Related Module: Admin Module

Acceptance Criteria: An Admin account can view all listings and users, remove any listing, suspend any account, and view a basic count-based activity report.

4.2 Non-Functional Requirements

The table below lists each non-functional requirement with a unique ID, matching the format used for functional requirements in Section 4.1, so each can be tracked and verified individually.

ID	Feature	Description	Priority
NFR-001	Performance — Response Time	Listing search and browse results must return within 3 seconds under normal load for a capstone-scale demo dataset.	High
NFR-002	Performance — Concurrent Users	The system must support at least [X] concurrent users without noticeable slowdown during demonstration.	Medium
NFR-003	Security — Password Storage	All passwords must be stored using a salted hash (e.g., bcrypt); plaintext passwords must never be stored or logged.	High
NFR-004	Security — Access Control	All non-public API endpoints must require session/token-based authentication, and actions must be restricted by role	High

		(Buyer/Seller/Admin) via role-based access control.	
NFR-005	Security — Input Validation	All user-submitted input must be validated and sanitized server-side to prevent SQL injection and cross-site scripting (XSS).	High
NFR-006	Reliability — Data Consistency	Reservation state transitions (Pending → Confirmed → Completed/Cancelled/Expired) must remain consistent even when multiple requests target the same listing concurrently.	High
NFR-007	Usability	The core reserve-an-item flow must be completable in 5 clicks or fewer from the listing detail page.	Medium
NFR-008	Maintainability	The system must maintain a clear separation between front-end, back-end, and database layers, consistent with the architecture in Section 3.1, so components can be modified independently.	Medium

Expanded detail per requirement, including related module and acceptance threshold:

NFR-001 — Performance — Response Time *(Priority: High)*

Listing search and browse results must return within 3 seconds under normal load for a capstone-scale demo dataset.

Related Module: Platform-wide

Acceptance Threshold: Search/browse requests measured via browser dev tools complete in 2 seconds or less against a demo dataset of roughly 50 listings.

NFR-002 — Performance — Concurrent Users *(Priority: Medium)*

The system must support at least [X] concurrent users without noticeable slowdown during demonstration.

Related Module: Platform-wide

Acceptance Threshold: The system remains responsive with at least [X] simultaneous browser sessions active during a live demo.

NFR-003 — Security — Password Storage *(Priority: High)*

All passwords must be stored using a salted hash (e.g., bcrypt); plaintext passwords must never be stored or logged.

Related Module: Authentication Module

Acceptance Threshold: Inspecting the users table directly shows only hashed values; no plaintext password appears anywhere in the database or logs.

NFR-004 — Security — Access Control *(Priority: High)*

All non-public API endpoints must require session/token-based authentication, and actions must be restricted by role (Buyer/Seller/Admin) via role-based access control.

Related Module: Authentication Module

Acceptance Threshold: Requests to protected endpoints without a valid token are rejected (401); a Buyer attempting an Admin-only action is rejected (403).

NFR-005 — Security — Input Validation *(Priority: High)*

All user-submitted input must be validated and sanitized server-side to prevent SQL injection and cross-site scripting (XSS).

Related Module: Platform-wide

Acceptance Threshold: Submitting script tags or SQL special characters in any form field is safely rejected or escaped, never executed or stored raw.

NFR-006 — Reliability — Data Consistency (*Priority: High*)

Reservation state transitions (Pending → Confirmed → Completed/Cancelled/Expired) must remain consistent even when multiple requests target the same listing concurrently.

Related Module: Reservation Module

Acceptance Threshold: Two simultaneous reservation attempts on the same listing result in exactly one successful reservation, never both.

NFR-007 — Usability (*Priority: Medium*)

The core reserve-an-item flow must be completable in 5 clicks or fewer from the listing detail page.

Related Module: Listing/Reservation UI

Acceptance Threshold: A first-time test user can complete a reservation in 5 clicks or fewer without being given instructions.

NFR-008 — Maintainability (*Priority: Medium*)

The system must maintain a clear separation between front-end, back-end, and database layers, consistent with the architecture in Section 3.1, so components can be modified independently.

Related Module: Platform-wide (Architecture)

Acceptance Threshold: Front-end, back-end, and database code are organized in clearly separated project directories, with no direct database queries issued from UI code.

4.2.1 Performance Requirements

NFR-001 and NFR-002 (above) govern response time and concurrent-user capacity. These are the requirements most likely to be checked live during the defense demo, so keep the demo dataset small enough to comfortably meet NFR-001.

4.2.2 Safety Requirements

Not applicable — the system does not control physical hardware or safety-critical processes.

4.2.3 Security Requirements

NFR-003, NFR-004, and NFR-005 (above) cover password storage, access control, and input validation respectively. Together these satisfy the security-implementation expectations detailed further in Section 9.

4.2.4 Software Quality Attributes

NFR-006 (Reliability), NFR-007 (Usability), and NFR-008 (Maintainability) cover the remaining quality attributes relevant to ThriftFinder's scope.

4.3 Requirement Prioritization and Traceability

Requirements are grouped into two implementation passes so the team always knows what "done" means at any point in the schedule.

4.3.1 Pass 1 — MVP

ID	Feature	Description	Priority
FR-001	Account Registration & Login	Users register/log in as Buyer or Seller (or both) using email and password. Includes email verification and password reset.	High

FR-00 2	User Profile	Users maintain a profile with name, contact info, profile photo, and view their own listing/reservation history.	High
FR-00 3	Create/Edit/Delete Listing	Sellers post secondhand items with title, category, condition, price, description, and up to 5 photos. Sellers can edit or remove active listings.	High
FR-00 4	Browse & Search Listings	Buyers browse all active listings and search/filter by keyword, category, price range, and condition.	High
FR-00 5	Listing Detail View	Buyers view full listing details: photos, description, price, seller info, and reservation status.	High
FR-00 6	Reserve Item	A buyer reserves an available item for a limited holding period (e.g., 24-48 hours) pending pickup/payment arrangement.	High
FR-00 7	Reservation Status Tracking	Reservations move through defined states: Pending → Confirmed → Completed, or Cancelled / Expired. Both parties can view current status.	High

4.3.2 Pass 2 — Enhancements

ID	Feature	Description	Priority
FR-008	Automatic Expiry	A reservation that is not confirmed within the holding period automatically expires and the listing becomes available again. Implementation note: checked lazily (i.e., evaluated against the current time whenever the listing/reservation is read or acted on) rather than via a scheduled cron job, since the free hosting tier's cron only runs once per day with imprecise timing — unsuitable for a time-sensitive expiry window.	Medium
FR-009	Buyer–Seller Chat	Once a reservation is confirmed, a buyer and seller can message each other in a real-time in-app chat thread to arrange handover details. Implemented using Supabase Realtime (Postgres change subscriptions) — no separate chat service required, and free-tier limits (200 concurrent connections, 2M messages/month) comfortably cover capstone-scale usage.	Medium
FR-010	Notifications	Users receive in-app (and/or email) notifications for reservation created, confirmed, cancelled, or expired events.	Medium

FR-01 1	Ratings & Reviews	After a completed reservation, buyer and seller can rate/review each other.	Low
FR-01 2	Admin Moderation Panel	Admins can view/remove listings and users that violate platform rules, and view basic platform activity reports.	Medium

4.3.3 Out of Scope (explicitly not built)

- Online/in-app payment processing or escrow (handover and payment are arranged directly between buyer and seller)
- Delivery, courier, or shipping logistics integration
- Native mobile applications (Android/iOS) — the system is a responsive web application only
- AI-based recommendation or pricing-suggestion engine
- Multi-language / multi-currency support
- Third-party marketplace integrations (e.g., Facebook Marketplace)

4.3.4 Traceability Matrix

Maps each requirement to its implementing module and forward to the specific Section 5 model that establishes its design, giving a preview/overview of what Section 5 will contain. (Each requirement's Source is documented in Section 4.1/4.2.) Fill in Test Case IDs once the test plan (Section 12) is written.

Req. ID	Module	Section 5 Reference

FR-001	Authentication Module	5.1 Use Case Diagram; 5.3 System Architecture
FR-002	User Management Module	5.1 Use Case Diagram; 5.2 ER Diagram
FR-003	Listing Management Module	5.1 Use Case Diagram; 5.2 ER Diagram
FR-004	Listing Management Module	5.1 Use Case Diagram; 5.4 Data Flow Diagram
FR-005	Listing Management Module	5.1 Use Case Diagram
FR-006	Reservation Module	5.1 Use Case Diagram; 5.4 Data Flow Diagram
FR-007	Reservation Module	5.2 ER Diagram; 5.4 Data Flow Diagram
FR-008	Reservation Module	5.4 Data Flow Diagram
FR-009	Communication Module	5.2 ER Diagram; 5.3 System Architecture
FR-010	Communication Module	5.3 System Architecture
FR-011	Reputation Module	5.2 ER Diagram
FR-012	Admin Module	5.1 Use Case Diagram; 5.3 System Architecture
NFR-001	Platform-wide	5.3 System Architecture
NFR-002	Platform-wide	5.3 System Architecture
NFR-003	Authentication Module	5.3 System Architecture
NFR-004	Authentication Module	5.3 System Architecture

NFR-005	Platform-wide	5.3 System Architecture
NFR-006	Reservation Module	5.2 ER Diagram; 5.4 Data Flow Diagram
NFR-007	Listing/Reservation UI	5.1 Use Case Diagram
NFR-008	Platform-wide (Architecture)	5.3 System Architecture

4.4 Business Rules

- A listing can have at most one active (Pending or Confirmed) reservation at a time; further reservation attempts are blocked while one is active.
- The reservation holding period is [24/48 — team to finalize] hours from the time a reservation is placed.
- A reservation not confirmed by the seller within the holding period automatically expires (see FR-008), and the listing becomes reservable again.
- Only the seller who owns a listing may edit or delete it; only the buyer who placed a reservation (or the listing's seller) may cancel it.
- A user must verify their email address before creating a listing or placing a reservation.
- Admins may remove any listing or suspend any account that violates platform rules, overriding normal ownership restrictions.
- Item price must be a positive number; listings without at least one photo cannot be published.

- A buyer may cancel their own Pending reservation at any time before it is confirmed; once Confirmed, cancellation requires mutual agreement (communicated via the chat in FR-009).
- A user suspended by an Admin cannot create listings, place reservations, or send messages until reinstated.

4.5 External Interface Requirements

4.5.1 User Interfaces

Responsive web UI covering: landing/browse page, listing detail page, create/edit listing form, reservation flow, user profile/dashboard, and admin panel.

4.5.2 Hardware Interfaces

None beyond standard client device camera/storage for uploading listing photos.

4.5.3 Software Interfaces

- Database management system: Supabase (managed PostgreSQL, free tier)
- Authentication: Supabase Auth (email/password, free tier)
- Real-time messaging: Supabase Realtime (Postgres change subscriptions, powers the buyer–seller chat in FR-009)
- Email service (for notifications): [specify, e.g., Supabase's built-in email or a free SMTP provider such as Resend/SendGrid free tier]
- Image storage: Supabase Storage (free tier, bundled with the database plan)

4.5.4 Communications Interfaces

HTTPS for all client-server communication; REST API (JSON) between front end and back end.

